

JEE ADVANCED - 6 | PAPER - 1

JEE 2022

Date	Maximum Marks	Duration
16th January, 2022	186	3 Hours

General Instructions

- The question paper consists of 3 Parts (Part I: **Physics**, Part II: **Chemistry**, Part III: **Mathematics**).
Each Part has **two** sections (Section 1 & Section 2).
- Section 1** contains 2 types of questions [Type A, Type B].
Type A contains **8 Multiple Correct Answers Type Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.
Type B contains **2 Paragraphs containing 2 Questions** on each paragraph. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.
- Section 2** contains **6 Numerical Value Type Questions**. The answer to each of the question is a numerical value. For each question, enter the correct numerical value of the answer. **If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled. (Example: 6, 81, 1.50, 3.25)**
- For answering a question, an ANSWER SHEET (OMR SHEET) is provided separately. Please fill your **Test Code**, **Roll No.** and **Group** properly in the space given in the ANSWER SHEET.

Name of the Candidate (In CAPITALS) :

Roll Number :

OMR Bar Code Number :

Candidate's Signature : Invigilator's Signature

MARKING SCHEME

SECTION - 1 | Type - A

- This section consists of **Eight (08) Questions**. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:
 - Full Marks:** +4 If only (all) the correct option(s) is(are) chosen
 - Partial Marks:** +3 If all the four options are correct but **ONLY** three options are chosen
 - Partial Marks:** +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct
 - Partial Marks:** +1 If two or more options are correct but **ONLY** one option is chosen, and it is a correct option
 - Zero Mark:** 0 if none of the options is chosen (i.e. the question is unanswered)
 - Negative Marks:** -1 In all other cases.

SECTION - 1 | Type – B

- This section has **TWO (02) Paragraphs containing TWO (02) Questions** on each paragraph. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:
 - Full Marks:** +3 If **ONLY** the correct option is chosen
 - Zero Marks:** 0 If none of the options is chosen (i.e. the question is unanswered)
 - Negative Marks:** -1 In all other cases.

SECTION 2

- This section has **SIX (06) Questions**. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer. **If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled. (Example: 6, 81, 1.50, 3.25)**
- Answer to each question will be evaluated according to the following marking scheme:
 - Full Marks:** +3 If **ONLY** the correct numerical value is entered;
 - Zero Marks:** 0 In all other cases.

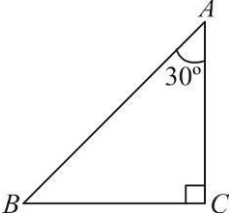
PART I : PHYSICS

62 MARKS

SECTION 1 | Type A

This section consists of Eight (08) Questions. Each question has FOUR options. ONE OR MORE THAN ONE of these four option(s) is(are) correct answer(s).

- A ray of light is incident on face AB of a right-angled prism ABC with $\angle A = 30^\circ$. The face BC of the prism is parallel to the horizontal. The refractive index of the prism is $\sqrt{2}$. The surrounding medium is air. Assume that after refraction at face AB , the ray always falls on face AC . Take $\sin 15^\circ = 0.26$. Which of the following statements is(are) true?



(A) If the incident ray is travelling at an angle 15° above the horizontal before it enters the prism through face AB , it emerges out of face AC horizontally

(B) If the incident ray is travelling vertically before it enters the prism through face AB , it undergoes total internal reflection at face AC

(C) If the incident ray is travelling horizontally before it enters the prism through face AB , it undergoes total internal reflection at face AC

(D) If the incident ray is travelling normally to face AB before it enters the prism through face AB , it emerges out of face AC at an angle 45° below the horizontal
- Two long insulating wires are placed parallel to each other in the x - y plane, along the lines $x = -a$ and $x = a$ respectively. The wires are uniformly charged with linear charge densities $-\lambda$ and λ respectively. Which of the following statements is(are) true?

(A) At any point on the z axis, the electric field is in the $-x$ direction

(B) The y - z plane is an equipotential surface

(C) The electric field at the point $(x, 0, 0)$ [$0 < x < a$] is $\left[\frac{\lambda x}{\pi \epsilon_0 (a^2 - x^2)} \right] (-\hat{i})$

(D) The work done in taking a point charge $+q$ from $(0, 0, 0)$ to $\left(\frac{a}{2}, 0, 0\right)$ is $\frac{\lambda q}{2\pi \epsilon_0} \log_e 3$
- A series RLC circuit consisting of an unknown resistance R , a capacitor of capacitance $C = 20\mu F$ and a coil of inductance $L = 0.5H$ is connected to an AC voltage source of angular frequency $\omega = 100\text{ rad/s}$. It is found that the current in the circuit leads the source voltage by a phase angle of $\frac{\pi}{4}$ and the average power dissipated in the resistance is $90W$. Which of the following statements is(are) true?

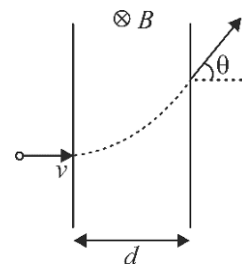
(A) The rms value of the current in the circuit is $\frac{1}{\sqrt{5}} A$

(B) The power factor is $\frac{1}{\sqrt{2}}$

(C) The amplitude of the voltage source is $180V$

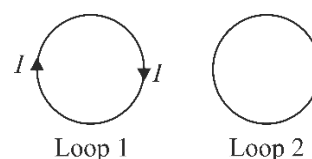
(D) The current in the circuit lags behind the voltage source by a phase angle of $\frac{\pi}{4}$ if the angular frequency of the source is changed to 1000 rad/s

4. A particle of mass m carrying charge $+q$ enters a region with a uniform magnetic field of intensity B directed into the plane of the paper as shown. The region lies between two parallel planes separated by a distance d . The velocity of the particle just before it enters the region is perpendicular and has magnitude v . The particle leaves the region after turning through an angle θ (in radian). Which of the following statements is(are) true?



- (A) The kinetic energy of the particle reduces by $\frac{1}{2}mv^2 \sin^2 \theta$ in passing through the region
- (B) $\sin \theta = \frac{qBd}{mv}$
- (C) The time spent by the particle inside the region is $\frac{m\theta}{qB}$
- (D) The force experienced by the particle while it is inside the region is constant in magnitude but varying in direction

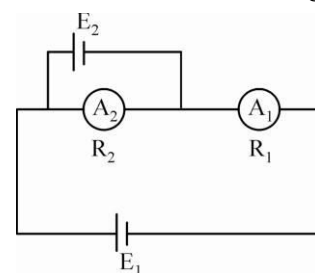
5. Two identical wire loops are placed in the same plane as shown and a clockwise current is made to flow through Loop 1. Loop 2 is moved in different ways as described, but it always remains in the same plane as Loop 1. Assume that the current in Loop 1 is kept constant. Which of the following statements is(are) true?



- (A) If Loop 2 is moved away from Loop 1, a clockwise current is induced in Loop 2
- (B) If Loop 2 is moved away from Loop 1, an anti-clockwise current is induced in Loop 2
- (C) If Loop 2 is moved closer to Loop 1, a clockwise current is induced in Loop 2
- (D) If Loop 2 is moved along a circular path around the centre of Loop 1, no current is induced in Loop 2
6. Consider two Bohr orbits A and B in the Hydrogen atom with principal quantum number $n = n_1$ and $n = n_2$ [$n_1 > n_2 > 1$], and radii r_1 and r_2 respectively. For the electron in the two orbits, let the angular momentum be L_1 and L_2 , and let the total energy be E_1 and E_2 . If the electron in orbit A transitions to the immediately lower orbit, it emits a photon of wavelength λ_1 and if the electron in orbit B transitions to the immediately lower orbit, it emits a photon of wavelength λ_2 . Which of the following options is(are) true?

- (A) $r_1 > r_2$ (B) $L_1 > L_2$ (C) $E_1 < E_2$ (D) $\lambda_1 > \lambda_2$

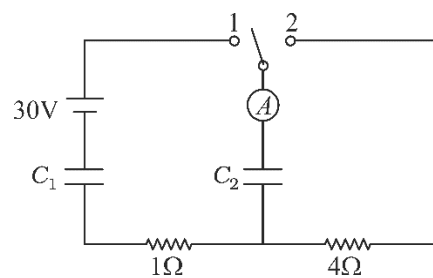
7. Two ideal batteries and two ammeters are arranged as shown in the figure. Which of the following statements is(are) true?



- (A) Readings of both ammeters can be same if $E_1 > E_2$
- (B) Readings of both ammeters can be same if $E_2 > E_1$ provided $R_2 > R_1$
- (C) Readings of both ammeters can be same if $E_2 > E_1$ provided $R_2 < R_1$
- (D) If $E_2 > E_1$, then current in ammeters will flow in opposite directions

8. In the circuit shown, all capacitors are initially uncharged. The capacitance of the capacitors are $C_1 = 2\mu\text{F}$ and $C_2 = 3\mu\text{F}$. The switch is closed at position 1 and kept closed until the current in the ideal ammeter becomes negligible. Which of the following statements is(are) true?

- (A) The potential difference across C_1 is 18V
 (B) The charge on C_2 is $36\mu\text{C}$
 (C) The heat dissipated in the 1Ω resistance is $540\mu\text{J}$
 (D) Now if the switch is shifted to position 2, the initial current registered by the ammeter is 3A



SPACE FOR ROUGH WORK

SECTION 1 | Type B

This section has TWO (02) Paragraphs containing TWO (02) Questions on each paragraph. Each question has FOUR options. ONLY ONE of these four options is the correct answer.

Paragraph for Q-9 to Q-10

In a Young's double slit experiment, separation between the slits is 0.1 mm and slits are illuminated by a monochromatic light source placed symmetrically behind the slits. It is observed that a point P on the screen at a distance 4.8 cm from the central maxima is the 6^{th} maxima from the central maxima.

Now, if the screen is moved towards the slits by a distance 0.5 m , the point P becomes the 8^{th} maxima from the central maxima.

9. The wavelength of the light used in the experiment is :
 (A) 200 nm (B) 300 nm (C) 400 nm (D) 600 nm
10. In the final situation, the distance between the central maxima and the closest point of complete darkness is:
 (A) 0.5 mm (B) 1 mm (C) 2 mm (D) 3 mm

Paragraph for Q-11 to Q-12

A heater consists of a metal rod. The variation of the resistance of the rod with temperature is described by the equation: $\frac{dR}{dT} = R_0\alpha$, where R_0 is the resistance of the rod at temperature $T_0 = 480\text{ K}$, α is a constant, R is the resistance of the rod in ohm and T is temperature of the rod in Kelvin.

The rod has very high thermal conductivity, so its entire volume always remains at the same temperature. The heat lost per unit time by the rod from its surface to the surroundings is proportional to $(T - T_s)$, where T is the temperature of the rod and $T_s = 300\text{ K}$ is the temperature of the surroundings, which remains constant.

When a potential difference $V_0 = 100\text{ Volts}$ is applied across the ends of the rod, its equilibrium temperature is found to be $T_0 = 480\text{ K}$, and when a potential difference $V = 120\text{ Volts}$ is applied across the ends of the rod, its equilibrium temperature is found to be $T = 540\text{ K}$.

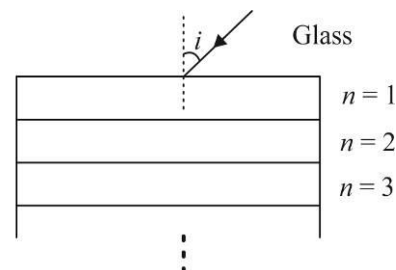
11. The value of α is (in K^{-1}) :
 (A) 6.67×10^{-4} (B) 2.67×10^{-3} (C) 1.33×10^{-3} (D) 3.33×10^{-4}
12. If the final current through the rod with $V_0 = 100\text{ Volts}$ is 4.5 A , the final current through it with $V = 120\text{ Volts}$ is :
 (A) 4.05 A (B) 5.00 A (C) 5.55 A (D) 6.17 A

SPACE FOR ROUGH WORK

SECTION 2

This section has SIX (06) Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled. (Example: 6, 81, 1.50, 3.25)

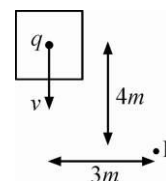
- A slab is made out of $N = 30$ layers of different transparent materials such that the refractive index of the n^{th} layer is $\mu_n = \left(\frac{3}{2} - \frac{n}{61}\right)$. All the interfaces between layers are parallel to each other. This slab is placed in contact with glass of refractive index 1.5 and a light ray travelling in the glass is incident on the slab at an angle of incidence $i = \sin^{-1}\left(\frac{4}{5}\right)$.



If the ray is able to penetrate to the m^{th} layer before undergoing total internal reflection, then $m = \underline{\hspace{2cm}}$.

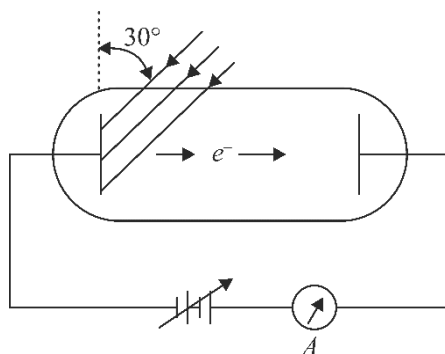
- A sample consists of nuclei of type A and nuclei of type B. Type A nuclei are radioactive and decay into type B nuclei which are stable. The half-life for the radioactive decay $A \rightarrow B$ is T . At $t = 0$, the sample contains A_1 nuclei of type A and B_1 nuclei of type B such that $\frac{A_1}{B_1} = \frac{1}{2}$. If at $t = t_0$, the sample contains A_2 nuclei of type A and B_2 nuclei of type B such that $\frac{A_2}{B_2} = \frac{1}{5}$, the ratio $\frac{t_0}{T}$ is equal to $\underline{\hspace{2cm}}$.

- A small elevator carrying a charge of $0.5C$ is moving down with a velocity of $5 \times 10^3 \text{ m/s}$. The elevator is $4m$ from the bottom and $3m$ horizontally from P as shown. What magnetic field (in μT) does it produce at point P ? $\underline{\hspace{2cm}}$.



- A coil of inductance $L = 10 \text{ mH}$ is connected in series with a resistance $R = 100 \Omega$ and a battery, and the switch is closed. The time after which current in the circuit is $\frac{3}{4}$ times its maximum value is $\underline{\hspace{2cm}}$ milliseconds. [Take $\log_e 2 = 0.7$]
- A large planar, thin insulating sheet is placed parallel to the horizontal and given a uniform positive charge density $\sigma = 2 \times 10^{-4} \text{ C/m}^2$. A small dipole of dipole moment $|\vec{p}| = (5 \times 10^{-7}) \text{ C m}$ is fixed above the sheet such that the dipole moment vector points in the vertically upward direction. The work done in slowly rotating the dipole such that its dipole moment vector now points in the vertically downward direction is $\underline{\hspace{2cm}}$ J. [Ignore gravity, take $\epsilon_0 = \frac{80}{9} \times 10^{-12} \text{ C}^2 / \text{Nm}^2$]

6. A metal plate of work function $\phi_0 = 2.7 \text{ eV}$ and surface area $A = 20 \text{ cm}^2$ is fixed inside a vacuum tube and light of wavelength $\lambda = 414 \text{ nm}$ and intensity $I = 500 \text{ W/m}^2$ is incident on the entire plate area making angle 30° with the plate as shown. The number of photoelectrons generated is $\frac{3}{100}$ times the number of photons striking the plate. A potential difference is applied across the tube such that the photoelectrons emitted are accelerated towards the facing plate. When the applied potential difference is increased, the maximum value that the current registered by the ideal ammeter reaches is _____ mA. [$hc = 1242 \text{ eV-nm}$, charge on an electron, $e = 1.6 \times 10^{-19} \text{ C}$]

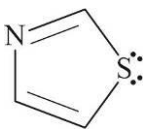


SPACE FOR ROUGH WORK

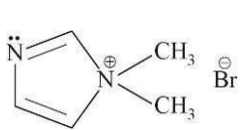
PART II : CHEMISTRY**62 MARKS****SECTION 1 | Type A**

This section consists of Eight (08) Questions. Each question has FOUR options. ONE OR MORE THAN ONE of these four option(s) is(are) correct answer(s).

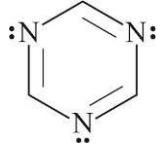
- Which of the following is/are correct?
 - zinc uranyl acetate is used as a laboratory reagent in the confirmatory test of the sodium ion
 - Nessler's reagent is used as a laboratory reagent in the confirmatory test of the ammonium ion
 - Potassium ferrocyanide on reaction with FeCl_3 solution form prussian blue
 - $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$ on heating strongly produce SO_2 and SO_3 gases along with Fe_2O_3 and water vapours
- According to Huckel rule, which is/are aromatic compound(s) ?



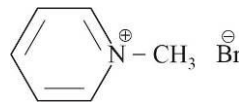
(A)



(B)



(C)



(D)
- Among the following complexes, the pair which shows ionization isomerism is/are :
 - $[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})\text{Cl}]\text{Br}_2$ and $[\text{Co}(\text{NH}_3)_4\text{Br}_2]\text{Cl} \cdot \text{H}_2\text{O}$
 - $[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})\text{Cl}]\text{Cl}_2$ and $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl} \cdot \text{H}_2\text{O}$
 - $[\text{Ru}(\text{ON})(\text{NO})_4(\text{OH})]^{2-}$ and $[\text{Ru}(\text{NO})(\text{NO})_4(\text{OH})]^{2-}$
 - $[\text{Co}(\text{en})_3][\text{Cr}(\text{CN})_6]$ and $[\text{Cr}(\text{en})_3][\text{Co}(\text{CN})_6]$
- When the intrinsic semiconductor germanium (Ge) is doped with phosphorus ?
 - there will be an increase in conductivity
 - it will be a n-type semiconductor
 - there will be a decrease in conductivity
 - it will be a p-type semiconductor
- Consider the following statement about Langmuir isotherm and identify correct statement(s) :
 - The free gas and adsorbed gas are in dynamic equilibrium
 - All adsorption sites are equivalent
 - The initially adsorbed layer can act as a substrate for further adsorption
 - The ability of a molecule to get adsorbed at a given site is independent of the occupation of neighboring site
- Consider the following statements about natural rubber and identify correct statement(s) ?
 - It is polymer of isoprene
 - On complete hydrogenation it produces ethylene propylene copolymer
 - On complete reductive ozonolysis it produces 4-keto pentanal as major product
 - The monomer units along the backbone of the carbon chain are in cis arrangement

7. Aspirin can be prepared by:
- (A) Reimer-Tiemann reaction of phenol
 - (B) Friedel crafts acylation on benzoic acid
 - (C) Esterification of phenol using salicylic acid
 - (D) Acylation of salicylic acid using acetic anhydride in the presence of phosphoric acid
8. Which of the following is/are green?
- (A) BaCl_2 salt in fire works
 - (B) Flame test for boron
 - (C) Malachite
 - (D) $\text{Be}_3\text{Al}_2(\text{SiO}_3)_8$, Cr^{3+} replacing a few Al^{3+} in octahedral sites

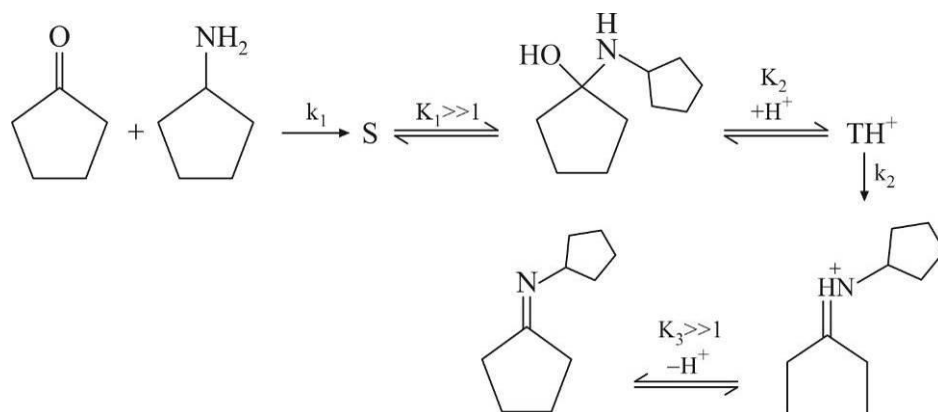
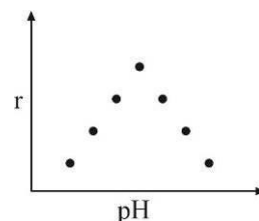
SPACE FOR ROUGH WORK

SECTION 1 | Type B

This section has TWO (02) Paragraphs containing TWO (02) Questions on each paragraph. Each question has FOUR options. ONLY ONE of these four options is the correct answer.

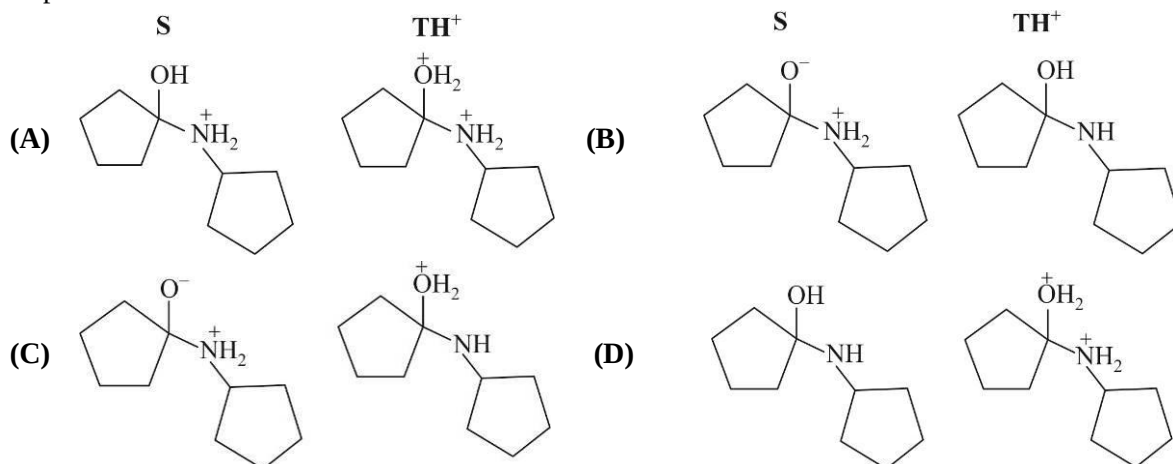
Paragraph for Q-9 to Q-10

The figure at the right demonstrate the change in rate of cyclopentanone and cyclopentylamine condensation with solution pH. The plot is followed by the mechanism of condensation.



k_1 and k_2 are rate constants K_1 , K_2 and K_3 are equilibrium constants.

9. Depict the correct structures of the intermediates S and TH^+ .

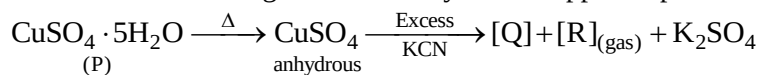


10. Which of the following is incorrect?

- (A) At low pH amine addition is limiting and called as RDS
 (B) At high pH, H_2O elimination is limiting and called as RDS
 (C) With pH decrease, amine is significantly protonated and simultaneously equilibrium K_2 is shifted to TH^+ side
 (D) With pH increase amine is significantly protonated and simultaneously equilibrium K_2 is shifted to TH^+ side

Paragraph for Q-11 to Q-12

Consider the following reactions of hydrated copper sulphate.



11. Which of the following is NOT correct ?
- (A) Coordination number of central atom in both (P) and (Q) is five
 - (B) Coordination sphere of (P) is square planar in shape
 - (C) Coordination sphere of (Q) is tetrahedral in shape
 - (D) Gas (R) when dissolved in sodium hydroxide undergoes disproportionation
12. The colour of light absorbed by an aqueous solution of CuSO_4 is :
- (A) Orange-red (B) Blue-green (C) Yellow (D) Violet
-

SPACE FOR ROUGH WORK

SECTION 2

This section has SIX (06) Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25)

1. How many isomers are possible for octahedral coordination compound having formula $\text{Fe}(\text{CN})_2 \cdot 4\text{KCN}$?
2. o-aminobenzaldehyde is heated with acetone in dil. NaOH to produce the major product (M). (M) on treatment with catalytic amount of H_2SO_4 produce another major product (N). What is percent of nitrogen in product (N) ? [Atomic mass : H = 1, C = 12, N = 14, O = 16]
3. How many lone pair of electrons are present in covalent compound I_2Cl_6 ?
4. 10gm of a mixture of ethyl chloride and ethanol are reacted with sodium to give 200 mL hydrogen at 27°C and 760mm pressure. What is the percentage of ethanol in the mixture ? [Atomic mass : H = 1, C = 12, O = 16]
5. For how many of the following d-block metals, standard reduction potential for M^{2+}/M electrode is positive? Ti, V, Cr, Mn, Fe, Co, Ni, Cu, Zn
6. How many of the following are sulphides ?
Galena, Cinnabar, Copper pyrites, Copper glance, Sphalerite, Cuprite, Siderite, Kaolinite

SPACE FOR ROUGH WORK

PART III : MATHEMATICS**62 MARKS****SECTION 1 | Type A**

This section consists of Eight (08) Questions. Each question has FOUR options. ONE OR MORE THAN ONE of these four option(s) is(are) correct answer(s).

- Let $y = f(x)$ is the solution of differential equation $\left(\frac{1}{x\sqrt{1-x^2}} - \frac{\sin^{-1}x}{x^2} - 2xy \right) dx = x^2 dy$ and $\lim_{x \rightarrow 0} f(x) = \frac{1}{6}$ then choose the correct option(s) :
 (A) $f(x)$ is an even function in the domain (B) $f(x)$ is an odd function in its domain
 (C) $f(x) + x^2 = 0$ has no real roots (D) $f(1) + f(-1) = \pi - 2$
- Let $f(x) = \int_x^{2x} (t \ln t - t) dt$ for all $x > 0$, Identify the correct statement(s) :
 (A) $f(x)$ is strictly increasing in $(0, e)$ and strictly decreasing in (e, ∞)
 (B) $f(x)$ is strictly decreasing in $\left(0, \frac{e}{2^{4/3}}\right)$ and strictly increasing in $\left(\frac{e}{2^{4/3}}, \infty\right)$
 (C) $f(x)$ has a local minima but no local maxima
 (D) $\lim_{x \rightarrow \infty} \frac{f'(x)}{x^2}$ is equal to zero
- The function $f(x) = \cos^{-1} \left(\frac{2[|\sin x| + |\cos x|]}{\sin^2 x + 2 \sin x + \frac{11}{4}} \right)$ is defined if x belongs to: (where $[.]$ denotes greatest integer function)
 (A) $\left[0, \frac{7\pi}{6}\right]$ (B) $\left[0, \frac{\pi}{6}\right]$ (C) $\left[\frac{11\pi}{6}, 2\pi\right]$ (D) $\left(\frac{7\pi}{6}, \frac{11\pi}{6}\right)$
- $\lim_{x \rightarrow \infty} \left(\frac{ax^2 + bx + c}{px^2 + qx + r} \right)^{lx^2 + mx + n}$, $(a, b, c, p, q, r, \ell, m, n \in \mathbb{R}^+)$
 (A) is equal to zero for $a < p$ (B) does not exist for $a > p$
 (C) is equal to one for $a = p$ (D) is equal to $e^{\frac{(b-q)m}{p}}$ for $a = p, \ell = 0$
- $\int \frac{\cos^{-1} \frac{x}{a}}{x^2} dx$ is equal to :
 (A) $-\frac{\pi}{2x} - \int \frac{\sin^{-1} \frac{x}{a}}{x^2} dx$ (B) $\frac{1}{a} \ln \left(\frac{a + \sqrt{a^2 - x^2}}{x} \right) - \frac{1}{x} \cos^{-1} \frac{x}{a} + C$
 (C) $\cos^{-1} \frac{x}{a} \ln x - x(\ln x - 1) + C$ (D) $\sqrt{a^2 - x^2} \ln x + x(\ln x - 1) + C$

6. Let $f(x) = \begin{cases} 3^x & ; x \leq 0 \\ (x+\lambda) & ; 0 < x < 2 \\ 3+(x-2)^2 & ; x \geq 2 \end{cases}$ is continuous for all $x \in R$, then which of the following is true:
- (A) Range of $f(x)$ is $(0, \infty)$ (B) $f(x) = 1$ has one solution
 (C) $f(x) = |f(x)|$ for all $x \in R$ (D) $f(x) = 1$ has two solutions
7. Person A randomly selects 4 distinct numbers from the set $\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$ and arrange them in descending order to form a 4 digit number and person B randomly selects 4 distinct numbers from set $\{1, 2, 3, 4, 5, 6, 7, 8\}$ and also arranges them in descending order to form a 4 digit number. Then which of the following is/are correct ?
- (A) probability that person A's 4 digit number is greater than person B's number is $\frac{{}^8C_3}{9C_4} + \frac{1}{2} \left(1 - \frac{1}{9C_4} \right)$
- (B) probability that person A and B have same 4 digit number is $\frac{1}{9C_4}$
- (C) probability that person A's 4 digit number is greater then person B's number is $\frac{2 \cdot {}^8C_3 + {}^8C_4 - 1}{2 \cdot 9C_4}$
- (D) probability that person A and B have same 4 digit number is $\frac{{}^8C_4}{9C_4}$
8. A and B are two $n \times n$ matrices such that $\det(A) \neq 0$, $A+B = (AB)^2$ and $BAB = A+I$. Choose the correct option(s).
- (A) $A^{-1} = (A^4 - I)$ (B) $B^5 - A^5 = A+B$
 (C) $A^9 = A^4 + A + I$ (D) $A^2B^2 = BA^2B$

SPACE FOR ROUGH WORK

SECTION 1 | Type B

This section has TWO (02) Paragraphs containing TWO (02) Questions on each paragraph. Each question has FOUR options. ONLY ONE of these four options is the correct answer.

Paragraph for Q-9 to Q-10

Let $f: R \rightarrow R$ be a function defined as $f(x) = 3x^5 - 25x^3 + 60x + 5$,

$$\text{and } g(x) = \begin{cases} \max\{f(t); -4 \leq t \leq x\} & ; -4 \leq x \leq 0 \\ \min\{f(t); 0 < t \leq x\} & ; 0 < x \leq 2 \\ f(x) - 16 & ; x > 2 \end{cases}$$

On the basis of given definitions of $f(x)$ and $g(x)$, answer the following questions.

9. Total number of points at which the graph of $y = g(x)$ is non differentiable in $[-4, \infty)$ is/are :
 (A) 1 (B) 2 (C) 0 (D) 4
10. If the equation $f(x) + \lambda = 0$ is having exactly three distinct real roots, then total number of possible integral values of ' λ ' are :
 (A) 20 (B) 21 (C) 42 (D) 40

Paragraph for Q-11 to Q-12

Let L be a straight line passing through $(1, 1, 1)$ which is parallel to $x + y + z = 1$ and intersect the line $x + 2y + z = 1, x + y + 2z = 3$

11. The line L is parallel to the line:
 (A) $\frac{x-1}{2} = \frac{y-1}{-3} = \frac{z-1}{1}$ (B) $\frac{x}{4} = \frac{y}{-6} = \frac{z}{2}$
 (C) $\frac{x}{5} = \frac{y}{-7} = \frac{z}{2}$ (D) $\frac{x}{4} = \frac{y}{-3} = \frac{z}{-1}$
12. The equation of line which intersect the line L and $\frac{x}{1} = \frac{y}{9} = \frac{z}{17}$ & parallel to $\frac{x}{1} = \frac{y}{2} = \frac{z}{3}$ is :
 (A) $\frac{x-1}{1} = \frac{y-1}{2} = \frac{z-2}{3}$ (B) $\frac{x-1}{2} = \frac{y-2}{4} = \frac{z-2}{3}$
 (C) $\frac{x}{7} = \frac{y}{14} = \frac{z}{21}$ (D) $\frac{x-1}{1} = \frac{y-1}{2} = \frac{z-1}{3}$

SPACE FOR ROUGH WORK

SECTION 2

This section has SIX (06) Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled. (Example: 6, 81, 1.50, 3.25)

- Let $f : R \rightarrow R$ be a continuous and differentiable function such that $f'(0) = 1$ and $f(x+2y) = f(x) + f(2y) + e^{x+2y}(x+2y) - xe^x - 2ye^{2y} + 4xy \forall x, y \in R$, then value of $f(1) - e$ is ____.
- If $\int_0^2 375x^5(1+x^2)^{-4} dx = 2^n$ then the value of n is :
- If A is a square matrix of order 2×2 such that $|A| = 3$, then sum of the infinite series $|A| + \left|\frac{1}{2}A\right| + \left|\frac{1}{4}A\right| + \left|\frac{1}{8}A\right| + \dots \infty$ is equal to :
- A coin is tossed 7 times. Each time a man calls head. If the probability that he wins the toss on more occasions is $\frac{a}{b}$ (where $a, b \in N$ in lowest terms and they are coprime) then $a+b$ is :
- If $f(x) = \lim_{n \rightarrow \infty} \frac{\sin(\pi + \pi x^{2n}) - (1 - x^{2n}) \tan(\pi x)}{1 + x^{2n} + x^{2n}(\sin(1+x) - \cos(\pi x))}$ and $g(x) = \begin{cases} x^2 + 5 & : x > 1 \\ 2x + b & : x \leq 1 \end{cases}$, then the value of $\left(\lim_{x \rightarrow 1} f(x) + b\right)$ is equal to (where $g(x)$ is continuous $\forall x \in R$).
- If $f(x)$ is a twice differentiable function & $f(0) = 2$, then $\lim_{x \rightarrow 0} \frac{\int_0^x (tf(x) + xf(t)) dt}{x^2}$ is equal to :

SPACE FOR ROUGH WORK